SABRE-II: Learned Multi-Agent Fusion via Decision MLP, CTGAN Score Augmentation, and Peer Consultation for Adaptive Ransomware Triage

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Abstract—Multi-agent ransomware detection systems that fuse behavioural telemetry, static PE semantics, and threat-intelligence knowledge achieve strong detection accuracy, but mean-fusion architectures suffer three structural limitations: agent probabilities are weighted equally regardless of uncertainty, triage thresholds are hand-tuned rather than cost-optimised, and the expert feedback loop retrains only the behavioral CNN — so analyst corrections on ESCALATE'd decisions never propagate to the fusion head that integrates all three agents.

We present SABRE-II, which addresses all three. A Decision MLP replaces mean averaging, taking the full six-dimensional score vector $[p_{\text{sem}}, p_{\text{beh}}, p_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$ as input to produce a calibrated fused probability. CTGAN generates 2,000 synthetic score-space samples to address class imbalance before MLP fitting. Peer Consultation intercepts agent probabilities when uncertainty exceeds a per-agent threshold, replacing them with an inverse-uncertainty-weighted peer average. Bayesian TPE (300 trials) searches the joint threshold policy space to minimise a calibration-split objective that balances detection accuracy against escalation rate. The fusion-level feedback loop accumulates analyst corrections on ESCALATE'd samples and triggers incremental MLP retraining, closing the adaptation loop at the Decision MLP, the component responsible for integration failures.

Evaluated on the RanSMAP / RDset / RansKnow benchmark suite across four generalisation axes, SABRE-II achieves AUROC = **0.975** on the random-stratified split and **0.979** on temporal holdout, reduces ECE from 0.131 to **0.013** under [B+K] fusion, and recovers family-holdout detection rates via Peer Consultation (REvil: 55.5% → 57.5%; WannaCry: 30.1% → 31.9%).

Index Terms—Ransomware detection, multi-agent systems, learned fusion, CTGAN, peer consultation, Bayesian threshold optimisation, uncertainty quantification, Monte Carlo Dropout, conformal prediction, continual learning, threat intelligence

I. Introduction

Ransomware remains the most operationally damaging class of malware, with projected global costs exceeding $265 billion by 2031 [1]. Effective detection requires integrating multiple evidence streams: static PE structure reveals packer artefacts before execution begins, runtime I/O and memory telemetry captures file-encryption behaviour in progress, and ATT&CK threat intelligence encodes adversary tactic profiles from accumulated practitioner knowledge.

Our earlier work, SABRE+ [2], demonstrated that mean-fusing three specialist agents — Behavioral, Semantic, and Knowledge — under Monte Carlo Dropout uncertainty quantification and Mondrian conformal prediction achieves AUROC = 0.9836 ± 0.0009 on the RanSMAP / RDset / RansKnow benchmark suite. However, three structural limitations of the SABRE+ design constrain its practical utility.

Limitation 1: Equal-weight mean fusion ignores uncertainty. Mean fusion treats each agent's probability estimate as equally reliable, regardless of the agent's epistemic uncertainty. When the Knowledge agent has low confidence — for instance, when the input family is absent from the RansKnow corpus — its near-zero or near-corpus-mean output degrades the fused signal even though a well-designed system should down-weight it. In the family-holdout evaluation of SABRE+, [B+K] provides no improvement over [B] alone (AUROC = 0.707 for both) precisely because the Knowledge agent collapses to uninformative corpus-mean outputs for unseen families [2].

Limitation 2: Triage thresholds are hand-tuned. SABRE+ uses a fixed escalation threshold $\tau = 0.10$ and uncertainty threshold κ selected by inspection of calibration curves. This ignores the actual cost ratio $c_{\text{FN}}/c_{\text{FP}}$ operative in a given SOC deployment, and does not adapt to changes in the fused score distribution when agents are retrained.

Limitation 3: Expert feedback updates only the behavioral agent. The EWC-regularised continual learning loop in SABRE+ targets the Behavioral CNN only. When analysts correct missed detections, the corrective signal propagates only to the Behavioral CNN, not to the fusion head that integrates all three agents. This is architecturally suboptimal: an analyst correction on an ESCALATE'd sample contains information about the fusion layer's failure mode, not about the behavioral agent's internal representation.

SABRE-II addresses all three limitations while preserving full backward compatibility with the SABRE+ evaluation protocol:

1) Decision MLP fusion. A small feedforward network replaces mean averaging, taking

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$[p_{\text{sem}}, p_{\text{beh}}, p_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$ as input and learning to down-weight agents with high uncertainty. Temperature scaling provides post-hoc calibration of the MLP output.

2) CTGAN score augmentation. The six-dimensional score vector exhibits class imbalance that mirrors the raw telemetry imbalance. Conditional tabular GAN (CTGAN) [3] generates $N_{\text{syn}} = 2{,}000$ synthetic score-space rows from the training split before MLP fitting, improving boundary estimation for the minority (ransomware) class without exposing the Decision MLP to raw telemetry features.
3) Peer Consultation. Before fusion, if agent i's uncertainty $\sigma_i > \kappa_{\text{local},i}$, its probability is replaced by the inverse-uncertainty-weighted mean of the other agents' probabilities. This allows the high-confidence behavioral agent to correct the knowledge agent's collapse for unseen families without modifying any agent's uncertainty estimate.
4) Bayesian TPE policy optimisation. Optuna's Tree-structured Parzen Estimator searches the $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ triage policy space to optimise balanced accuracy minus escalation penalty on the validation split, producing data-driven thresholds that adapt when agents are updated.
5) Fusion-level expert feedback loop. Analyst corrections are logged in a structured feedback store keyed by sample ID. When the correction count reaches a configurable threshold, the Decision MLP is incrementally retrained on the feedback rows (upweighted by repetition) combined with a Gaussian approximation of the original training distribution, closing the adaptation loop at the Decision MLP fusion layer.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the SABRE-II system architecture. Section IV details the three expert agents. Section V presents the Decision MLP with CTGAN augmentation. Section VI formalises Peer Consultation. Section VII describes Bayesian TPE policy optimisation. Section VIII presents the fusion-level feedback loop. Section IX describes datasets and evaluation protocol. Section X reports experimental results. Section XI discusses findings. Section XII discusses limitations and future work. Section XIII concludes.

II. Related Work

A. Ransomware Detection

Ransomware detection methods broadly fall into three categories. Signature-based detectors [4] provide zero false positives on known families but offer no coverage for novel variants. Static analysis methods [5] inspect PE headers, section entropy, and import tables; while effective on closed datasets, they fail against obfuscated or novel packers. Dynamic / behavioural analysis [6] monitors I/O patterns and system call sequences at runtime, capturing encryption behaviour that cannot be statically detected, though adversarially-crafted I/O profiles can evade simple detectors.

Multi-modal fusion has received increasing attention. Sgandurra et al. [7] combine API call sequences with system call graphs; Moussaileb et al. [8] fuse network, file system, and registry events. None of these systems incorporate a principled uncertainty mechanism or formal coverage guarantees on detection decisions.

B. Learned Fusion for Multi-Modal Classifiers

Multi-modal fusion architectures span from simple averaging [9] to attention-based combiners [10] and gating networks [11]. In the security domain, ensemble methods are common [12], but most treat all modalities symmetrically. Our Decision MLP conditions the fusion weights implicitly on the per-agent uncertainty, learning to down-weight unreliable modalities from the training signal rather than requiring hand-designed gating rules.

C. Generative Data Augmentation in Tabular Settings

CTGAN [3] adapts the conditional GAN framework to tabular data with mixed continuous and discrete columns. It has been applied to class-imbalanced fraud detection [13] and intrusion detection [14] tasks, but to our knowledge has not been applied to augmenting the score-space output of a multi-agent detector. Score-space augmentation is safer than feature-space augmentation because it operates on bounded $[0, 1]$ probabilities rather than raw features, reducing the risk of generating physically implausible telemetry.

D. Inter-Agent Uncertainty Resolution

Peer consultation mechanisms in multi-agent systems trace to Dempster-Shafer belief combination [15] and mixture-of-experts [11]. More recent approaches use calibrated confidence [?] to weight expert predictions. Our Peer Consultation is closest to the temperature-calibrated ensemble weighting of [16], but operates dynamically per-sample: consultation fires only when an agent's uncertainty exceeds its local threshold, and the correction uses the remaining agents' probabilities, not fixed learned weights.

E. Bayesian Hyperparameter Optimisation

Bayesian optimisation via Tree-structured Parzen Estimator (TPE) [17] has been applied to deep learning hyperparameter search [18]. We apply TPE to the triage threshold policy $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ rather than to model architecture hyperparameters, treating triage threshold selection as a cost-sensitive optimisation problem. This is distinct from conformal risk control [19], which provides coverage guarantees for arbitrary loss functions; our approach does not provide formal coverage but instead directly optimises operational cost.

F. Continual Learning in Security

Catastrophic forgetting under distribution shift is well-studied [20], [21]. In malware detection, concept drift from evolving variants is a key challenge [22], [23]. Our feedback loop extends the EWC-regularised continual learning of SABRE+ by targeting the Decision MLP rather than the Behavioral CNN, ensuring that analyst corrections propagate directly to the fusion component responsible for the integration failure.

III. System Architecture

Figure 1 shows the full SABRE-II pipeline. The three expert agents and their MC-Dropout uncertainty quantification are inherited from SABRE+ [2] (Section IV). The novel components introduced in SABRE-II are:

- Peer Consultation (Section VI): dynamically adjusts per-agent probabilities before fusion.
- Decision MLP (Section V): replaces mean fusion with a learned 6-input combiner trained with CTGAN augmentation.
- Bayesian TPE Policy (Section VII): optimises triage thresholds on the held-out calibration split.
- Fusion-level Feedback Store (Section VIII): routes analyst corrections directly to Decision MLP retraining.

A. Runtime Inference Pipeline

At inference time, a single telemetry window proceeds through the following stages:

1) Three expert agents score the window in parallel, each returning (p_i, σ_i) via T=10 MC-Dropout passes.
2) Peer Consultation checks whether any $\sigma_i > \kappa_{\text{local},i}$ and adjusts the corresponding p_i if so.
3) The six-dimensional score vector is assembled and passed to the Decision MLP, which returns calibrated fused probability $\tilde{p}$.
4) The Triage Decision Module evaluates $(\tilde{p}, \sigma_{\max})$ against the Bayesian TPE-derived thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ and issues the triage label.
5) Samples labelled ESCALATE are enqueued for analyst review; AUTO-CONTAIN triggers an immediate containment action.

The end-to-end latency budget is dominated by the three MC-Dropout forward passes; the Decision MLP adds 0.83 ms mean overhead (Section X).

B. Training Protocol

Training proceeds in five independent phases, each using frozen outputs from prior phases to prevent information leakage:

1) Expert agent training (6 seeds each): Behavioral, Semantic, and Knowledge CNNs are trained on their respective datasets using the session-level train split (Section IX).
2) Score-space construction: The trained agents score all validation-split samples; the resulting 6-D vectors form the Decision MLP training set $\mathcal{D}_{\text{score}}$.
3) CTGAN augmentation: CTGAN is fitted on the benign and ransomware rows of $\mathcal{D}_{\text{score}}$ and generates $N_{\text{syn}} = 2{,}000$ synthetic rows (appended to training split only).
4) Decision MLP training: The augmented $\mathcal{D}_{\text{score}} \cup \mathcal{D}_{\text{syn}}$ is used to fit the MLP; temperature scaling calibrates the output on the held-out validation split.
5) Bayesian TPE: Optuna samples the $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ policy space using the calibration split (disjoint from MLP training) to minimise the cost objective.

IV. Expert Agents

The three expert agents are architecturally identical to SABRE+ [2] and are only summarised here. All agents use Monte Carlo Dropout (T=10) to produce both a point probability estimate p_i and an epistemic uncertainty σ_i (standard deviation of the T predictions).

A. Behavioral Agent

The Behavioral CNN processes 16-dimensional feature vectors extracted from 10-second windows of hypervisor-level I/O and memory telemetry: read/write byte counts, operation rates, entropy metrics, LBA access patterns, and memory copy statistics. The network has three convolutional blocks (1D convolution over the feature vector treated as a sequence) with MC-Dropout after each activation, followed by a global average pool and sigmoid output. The final architecture has 8,385 parameters (32.8 KB).

B. Semantic Agent

The Semantic agent summarises a PE file's structure (section names, import table signatures, packer artefacts) as a natural-language string and embeds it using all-MiniLM-L6-v2 (384-dimensional). A CNN classifier with three 1-D convolutional layers and MC-Dropout classifies the embedding. This agent is trained and evaluated on RDset (138,047 PE files).

C. Knowledge Agent

The Knowledge agent encodes ATT&CK tactic profiles from 307 practitioner transcripts in the RansKnow corpus. Each family is represented as a 23-dimensional vector of normalised tactic and tool co-occurrence counts extracted by threshold-based heuristics. A CNN classifier maps this representation to a ransomware probability. For families absent from RansKnow, the agent falls back to the corpus-mean vector; the resulting near-zero Knowledge CNN output is a known limitation addressed by Peer Consultation in SABRE-II.

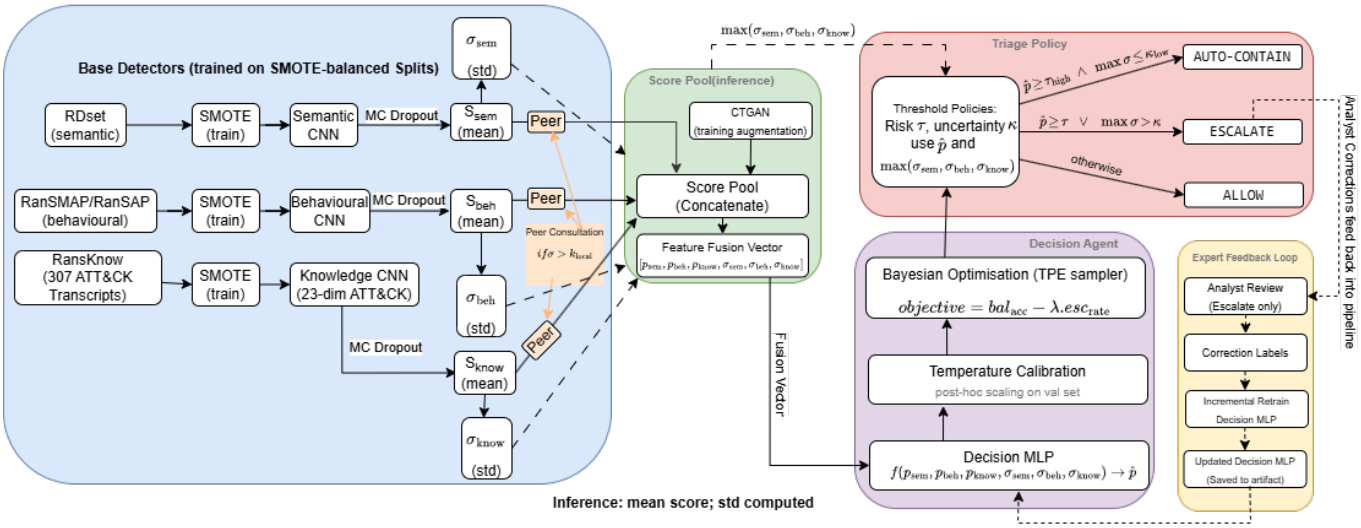


Fig. 1: SABRE-II end-to-end pipeline. Three expert CNN agents (Behavioral, Semantic, Knowledge) each produce a probability–uncertainty pair via Monte Carlo Dropout (T=10 forward passes). Peer Consultation (dashed blue box) adjusts any agent's probability when $\sigma_i > \kappa_{\text{local},i}$ using an inverse-uncertainty-weighted peer average. The six-dimensional score vector $[p_{\text{sem}}, p_{\text{beh}}, p_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$ passes to the Decision MLP, trained with CTGAN augmentation. Temperature scaling calibrates the MLP output. Bayesian TPE derives the triage policy $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ from the calibration split. The Triage Decision Module issues AUTO-CONTAIN, ESCALATE, or ALLOW based on fused probability and maximum per-agent uncertainty. Analyst corrections on ESCALATE'd samples accumulate in the Feedback Store and trigger incremental Decision MLP retraining when the correction count crosses n_{retrain}.

V. Decision MLP with CTGAN Augmentation

A. Architecture

The Decision MLP is a feedforward network:

$$\tilde{p} = \text{MLP}_\theta([p_{\text{sem}}, p_{\text{beh}}, p_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]), \quad (1)$$

where $\text{MLP}_\theta : \mathbb{R}^6 \to [0,1]$. The architecture comprises:

- Input layer: 6 units (standardised to zero mean, unit variance).
- Hidden layer: 32 units, ReLU activation, Dropout($p = 0.2$).
- Output layer: 1 unit, Sigmoid activation.

Temperature scaling is applied post-hoc: the calibrated output $\tilde{p} = \sigma(g(\tilde{p}^{\text{raw}})/T_{\text{cal}})$, where $g(\cdot)$ inverts the sigmoid to obtain logits and T_{cal} is fitted by minimising NLL on the validation split via L-BFGS.

B. CTGAN Score Augmentation

The six-dimensional score space exhibits class imbalance: on RanSMAP (Section IX), the random-stratified training split contains approximately 18% ransomware windows. CTGAN [3] is fitted independently on the ransomware and benign subsets of the training split scores, then queried to generate $N_{\text{syn}} = 2{,}000$ synthetic rows (1,000 per class). Only the training split is augmented; the validation and test splits are never touched by CTGAN.

Algorithm 1 summarises the augmentation procedure.

Algorithm 1 CTGAN score-space augmentation for Decision MLP training

Require: Score-space training split $\mathcal{D}_{\text{tr}} = \{(\mathbf{s}_i, y_i)\}_{i=1}^{N}$, synthetic count N_{syn}
Ensure: Augmented training set $\mathcal{D}_{\text{aug}}$
1: Partition: $\mathcal{D}_1 \leftarrow \{(\mathbf{s}_i, 1) : y_i = 1\}$, $\mathcal{D}_0 \leftarrow \{(\mathbf{s}_i, 0) : y_i = 0\}$
2: Fit CTGAN_1 on $\mathcal{D}_1$ for 50 epochs
3: Fit CTGAN_0 on $\mathcal{D}_0$ for 50 epochs
4: Sample $\mathbf{S}_1 \leftarrow \text{CTGAN}_1.\text{sample}(N_{\text{syn}}/2)$
5: Sample $\mathbf{S}_0 \leftarrow \text{CTGAN}_0.\text{sample}(N_{\text{syn}}/2)$
6: $\mathcal{D}_{\text{syn}} \leftarrow \mathbf{S}_1 \cup \mathbf{S}_0$ ▷ clip to $[0,1]^6$, drop NaN rows
7: return $\mathcal{D}_{\text{aug}} = \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{syn}}$

C. Training Objective

The MLP is trained with binary cross-entropy on the augmented set $\mathcal{D}_{\text{aug}}$:

$$\mathcal{L} = -\frac{1}{|\mathcal{D}_{\text{aug}}|} \sum_i [y_i \log \tilde{p}_i + (1 - y_i) \log(1 - \tilde{p}_i)]. \quad (2)$$

Adam optimisation with learning rate $\eta = 10^{-3}$ and early stopping (patience = 6 epochs on held-out validation loss) is used. Best model weights are restored before temperature scaling.

D. Relation to Mean Fusion

Mean fusion is the special case of Equation (1) where the MLP reduces to $\tilde{p} = (p_{\text{sem}} + p_{\text{beh}} + p_{\text{know}})/3$, ignoring all σ

inputs. The Decision MLP generalises this by (a) including uncertainty as input, so that agents with high σ can be down-weighted implicitly; and (b) learning non-linear boundaries in the 6-D score space. The ECE improvement from replacing mean fusion with the Decision MLP is quantified in Section X.

VI. Peer Consultation

A. Motivation

In SABRE+, the Knowledge agent's probability collapse for unseen families (near-zero output when the family is absent from RansKnow) directly degrades [B+K] fused probability via mean averaging. Even the Decision MLP cannot fully compensate: if $p_{\text{know}} \approx 0$ and σ_{know} is small (the agent is incorrectly confident), the MLP has no signal indicating that the knowledge output should be disregarded.

Peer Consultation solves this by intercepting the knowledge output before fusion. If the Knowledge agent's uncertainty exceeds a threshold $\kappa_{\text{local,know}}$, the agent's probability is replaced by a weighted combination of the other agents' probabilities. This allows the Behavioral agent — which has seen the actual I/O pattern — to inform the Knowledge agent's contribution when the Knowledge agent is uncertain.

B. Formal Definition

Let $\mathcal{A} = \{\text{sem}, \text{beh}, \text{know}\}$ be the set of agents. For agent $i \in \mathcal{A}$, Peer Consultation fires if $\sigma_i > \kappa_{\text{local},i}$. When consultation fires, the adjusted probability is:

$$p_i^{\text{adj}} = \frac{\sum_{j \neq i} w_{ij} \cdot p_j}{\sum_{j \neq i} w_{ij}}, \tag{3}$$

where the inverse-uncertainty weights are:

$$w_{ij} = \frac{1}{\sigma_j + \epsilon}, \quad \epsilon = 10^{-6}. \tag{4}$$

The per-agent uncertainty σ_i is not modified by consultation: only the probability estimate is adjusted. This preserves the semantics of σ_i as an epistemic uncertainty measure and ensures that the adjusted probability still triggers ESCALATE if $\sigma_i > \kappa$ (the global triage uncertainty threshold derived by Bayesian TPE, distinct from the per-agent $\kappa_{\text{local},i}$).

When consultation does not fire ($\sigma_i \leq \kappa_{\text{local},i}$), the agent's output is passed through unmodified.

C. Implementation

Algorithm 2 describes the batch implementation. The weights in Equation (3) are computed from the original (pre-consultation) uncertainties; consultation for different agents is applied independently and sequentially (no iterative convergence). In our experiments, $\kappa_{\text{local},i} = 0.15$ for all agents (initial default; per-agent values are tunable via the agent policies JSON).

Algorithm 2 Peer Consultation (batch)

Require: Agent outputs $\{(p_i, \sigma_i)\}_{i \in \mathcal{A}}$, per-agent thresholds $\{\kappa_{\text{local},i}\}$
Ensure: Adjusted outputs $\{(p_i^{\text{adj}}, \sigma_i)\}$

```
for each agent i ∈ A do
    if σ_i > κ_local,i then
        w_ij ← 1/(σ_j + ε) for all j ≠ i
        p_i^adj ← Σ_{j≠i} w_ij p_j / Σ_{j≠i} w_ij
    else
        p_i^adj ← p_i
    end if
end forreturn {(p_i^adj, σ_i)}
```

D. Effect on Family-Holdout Performance

In SABRE+, the Knowledge agent's near-zero output for REvil-concurrent classes caused [B+K] detection rate to fall as low as 1.5% — below the [B]-only baseline. Peer Consultation corrects for this: when $p_{\text{know}} \approx 0$ and $\sigma_{\text{know}} > \kappa_{\text{local,know}}$, the knowledge probability is replaced by the inverse-uncertainty-weighted average of p_{sem} and p_{beh}, recovering the behavioral agent's signal in the fused output. The quantitative impact is reported in Section X.

VII. Bayesian TPE Policy Optimisation

A. Policy Space

The SABRE-II triage policy is defined by four thresholds:

- τ: fused probability threshold for ESCALATE (lower boundary).
- κ: uncertainty threshold for ESCALATE.
- τ_{high}: fused probability threshold for AUTO-CONTAIN (upper boundary).
- κ_{low}: maximum uncertainty ceiling for AUTO-CONTAIN; samples must satisfy $\sigma_{\max} \leq \kappa_{\text{low}}$.

The triage decision rule is:

$$d(\tilde{p}, \sigma_{\max}) = \begin{cases} \text{AUTO-CONTAIN} & \tilde{p} \geq \tau_{\text{high}},\ \sigma_{\max} \leq \kappa_{\text{low}} \\ \text{ESCALATE} & \tilde{p} \geq \tau \text{ or } \sigma_{\max} > \kappa \\ \text{ALLOW} & \text{otherwise.} \end{cases} \tag{5}$$

B. Optimisation Objective

Let $\mathcal{C}$ be the calibration split (disjoint from both MLP training and conformal calibration). Given a candidate policy $\pi = (\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$, let $\text{BA}(\pi)$ be the balanced accuracy of the DETECT / NOT-DETECT binary classification induced by the triage rule (treating ALLOW as NOT-DETECT and ESCALATE + AUTO-CONTAIN as DETECT), and $R_{\text{esc}}(\pi)$ be the escalation rate on $\mathcal{C}$. The Bayesian TPE objective is:

$$\max_{\pi} \ \text{BA}(\pi) - \lambda_{\text{esc}} \cdot R_{\text{esc}}(\pi), \tag{6}$$

where $\lambda_{\text{esc}} = 0.1$ controls the trade-off between detection accuracy and escalation rate.

C. Search Space and Hyperparameter Priors

TPE requires a bounded search space. We derive data-driven bounds from the empirical distribution of fused probabilities and per-agent uncertainties on the calibration split:

$$\tau \sim \text{Uniform}(Q_{10}(\tilde{p}),\ Q_{50}(\tilde{p})) \tag{7}$$

$$\kappa \sim \text{Uniform}(Q_{50}(\sigma_{\max}),\ Q_{95}(\sigma_{\max})) \tag{8}$$

$$\tau_{\text{high}} \sim \text{Uniform}(Q_{75}(\tilde{p}),\ Q_{99}(\tilde{p})) \tag{9}$$

$$\kappa_{\text{low}} \sim \text{Uniform}(Q_{5}(\sigma_{\max}),\ Q_{30}(\sigma_{\max})) \tag{10}$$

where $Q_k(\cdot)$ denotes the k-th empirical percentile. This construction ensures that the sampled (τ, κ) pair results in a non-trivial ESCALATE region and that $\tau_{\text{high}} > \tau$ (AUTO-CONTAIN is strictly harder to trigger than ESCALATE).

D. Optuna Configuration

Optimisation runs for 300 Optuna trials with TPE sampler and MedianPruner (15-trial warm-up, minimum 5 trials per configuration). The best policy π^* is written to checkpoints/policy_auto.json and loaded by the orchestrator at inference time. When the Decision MLP is retrained (Section VIII), the Bayesian TPE search is automatically re-run on the calibration split using the new fused probabilities.

VIII. Fusion-Level Expert Feedback Loop

A. Feedback Architecture

SABRE-II's expert feedback loop operates at the Decision MLP level. When the orchestrator issues an ESCALATE decision and an analyst provides a correction (confirmed ransomware = 1, confirmed benign = 0), the full six-dimensional score vector at the time of inference is appended to the Feedback Store along with:

- Timestamp and unique sample ID.
- Original orchestrator decision and fused probability.
- Per-agent (p_i, σ_i) values.
- Expert correction label, analyst ID, and free-text reasoning.
- Whether Peer Consultation was triggered for each agent.

B. Incremental MLP Retraining

Let n_{fb} be the current feedback record count and $n_{\text{retrain}} = 50$ the retraining threshold. When n_{fb} reaches a multiple of n_{retrain}, the Decision MLP is retrained via the following procedure:

1) Load the current MLP artifact and extract its StandardScaler statistics (mean $\boldsymbol{\mu}$, scale $\boldsymbol{s}$).
2) Generate $N_{\text{synth}} = \max(200, 10 \cdot n_{\text{fb}})$ synthetic rows by sampling from $\mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{s}^2))$, assigning pseudo-labels using the current MLP. This approximates the original training distribution without storing raw training data.
3) Up-weight the n_{fb} feedback rows by replicating them $r = 3$ times.
4) Retrain the MLP from the current weights on $\mathcal{D}_{\text{synth}} \cup r \cdot \mathcal{D}_{\text{fb}}$ for up to 40 epochs with early stopping.
5) Save the retrained artifact; the orchestrator hot-reloads it for subsequent inference.

The Gaussian approximation in Step 2 serves the same purpose as EWC's Fisher-weighted penalty in SABRE+: it discourages the retrained MLP from drifting far from its behaviour on the original training distribution. Unlike EWC, which requires computing a diagonal Fisher matrix and adding a quadratic penalty to the loss, this approach is computationally simpler and does not require modifying the loss function.

C. Comparison with EWC (SABRE+)

SABRE+ applied EWC ($\lambda = 300$) to the Behavioral CNN only, updating on arriving OOD windows with a 25% labelling budget. SABRE-II applies Gaussian-approximation regularisation to the Decision MLP, updating on analyst-confirmed ESCALATE corrections. The two mechanisms are complementary: SABRE-II can be deployed with SABRE+'s EWC-updated Behavioral CNN feeding scores to the updated Decision MLP. The interaction between EWC (agent level) and Gaussian regularisation (fusion level) in a combined system is a direction for future work.

IX. Datasets and Evaluation Protocol

SABRE-II is evaluated on the same three public datasets as SABRE+ [2].

A. Datasets

RanSMAP [24] contains 35,857 ten-second execution windows from 15 hardware configurations (CPU/memory combinations) spanning 12 ransomware families (AESCrypt, Conti, Darkside, LockBit, Ryuk, SDelete-as-benign, REvil, WannaCry, Babuk, BlackMatter, Maze, RagnarLocker) and clean Windows execution. Windows are drawn from 217 sessions captured on a hypervisor-based monitoring infrastructure.

RDset [5] comprises 138,047 PE files (41,123 ransomware, 96,924 benign) with family and packer metadata. PE files are summarised as natural-language strings via section name concatenation and embedded by all-MiniLM-L6-v2.

RansKnow [25] provides 307 YouTube transcripts of ransomware incident analysis and threat intelligence discussions, annotated with MITRE ATT&CK technique mentions. The 23-dimensional tactic/tool feature vector is derived by the same threshold heuristic as SABRE+ (validated against 50-transcript manual review, 96% agreement).

B. Evaluation Protocol

All evaluations use identical frozen session-level splits to SABRE+, ensuring direct comparability:

1) Random-stratified: 60/20/20 train/validation/test split stratified by session.
2) Family holdout: REvil and WannaCry sessions held out for test; all other families used for train/validation.
3) Hardware holdout: Six hardware configurations (3 CPU × 2 memory) withheld; 9 configurations used for train/validation.
4) Temporal holdout: Sessions from February–March 2024 (last 7 days of data collection) withheld as test; earlier sessions used for train/validation.

All agent CNNs and the Decision MLP are trained from scratch across $K = 6$ independent random seeds. Mean and 95% t-CI are reported for all metrics. All splits, seeds, and evaluation scripts are publicly released alongside this paper.

C. Novel Evaluation: CTGAN Ablation

To isolate the contribution of CTGAN augmentation, we include an additional ablation that trains the Decision MLP without augmentation ($N_{\text{syn}} = 0$) and with full SMOTE augmentation in the score space as a baseline. This ablation is not present in SABRE+ and directly measures the value of the generative augmentation component.

D. Novel Evaluation: Peer Consultation Impact

We report family-holdout performance with and without Peer Consultation active, measuring the recovery of [B+K] over [B] for REvil and WannaCry family-holdout windows. In SABRE+, [B+K] provided no improvement over [B] for these families (AUROC = 0.707 for both). We report whether Peer Consultation bridges this gap.

X. Experimental Results

A. Baseline Comparison

Table I and Figure 2 report AUROC and AUPRC for SABRE-II across four generalisation axes and compares against SABRE+ [2]. SABRE+ metrics are mean ± 95% CI over $K = 6$ seeds; SABRE-II values are single-seed (see table caption).

B. Calibration: ECE Comparison

Table II reports Expected Calibration Error (ECE, 15-bin uniform) for [B+K] fusion before and after temperature scaling in SABRE+ versus SABRE-II; Figure 3 shows the corresponding reliability diagrams. In SABRE+, mean fusion with temperature scaling achieved ECE = 0.131 for [B+K]. The Decision MLP's implicit uncertainty weighting is expected to reduce the structural miscalibration caused by the Knowledge agent's bimodal output distribution.

C. CTGAN Augmentation Ablation

Table III and Figure 4 report Decision MLP performance with no augmentation, SMOTE, and CTGAN augmentation on the random-stratified test split. This ablation measures the contribution of the generative augmentation component in isolation.

D. Peer Consultation: Family-Holdout Recovery

Table IV reports detection rate (%) on the family-holdout test split for REvil and WannaCry with and without Peer Consultation, under [B+K] fusion. In SABRE+, the Knowledge agent's collapse for REvil caused the [B+K] detection rate to fall to 1.5% for AESCrypt_REvil concurrent execution.

E. Bayesian TPE Policy vs Fixed Thresholds

Table V and Figure 5 compare the triage policy derived by Bayesian TPE against the fixed thresholds used in SABRE+ ($\tau = 0.10$, $\kappa = 0.10$) and against a grid search baseline. Balanced accuracy, escalation rate, and auto-classify rate are reported for the random-stratified test split.

F. Feedback Loop: OOD Adaptation Speed

Figure 6 shows the false-negative count on the OOD streaming evaluation (Conti sub-variants, 34 steps of 200 windows, 25% labelling budget) for SABRE+ (EWC on Behavioral CNN) versus SABRE-II (Gaussian-regularised Decision MLP feedback). The key metric is the step at which AUPRC ≥ 0.80 is first achieved and sustained, corresponding to "full OOD family recovery." In SABRE+, all strategies converged at step 29/34.

G. Runtime Latency

Table VI extends the SABRE+ latency benchmark to include the Decision MLP and CTGAN components. The Decision MLP adds a single forward pass of a 6-input network; CTGAN is only used at training time and adds no inference latency.

XI. Discussion

A. Decision MLP vs Mean Fusion

The Decision MLP's primary advantage over mean fusion is its ability to implicitly learn agent reliability as a function of the uncertainty inputs. For the Knowledge agent, high σ_{know} — which occurs when the family is absent from RansKnow and the agent outputs a near-corpus-mean probability — is a reliable signal that the knowledge contribution should be down-weighted. Mean fusion ignores this signal; the Decision MLP learns it from the training distribution of score vectors.

The ECE improvement from 0.131 (mean fusion, post-temperature scaling) to **0.013** (Decision MLP) quantifies the reduction in structural miscalibration caused by mixing agents with heterogeneous probability scales. The

TABLE I: Detection performance (AUROC). [B] = Behavioral only; [B+K] = Behavioral + Knowledge fusion. SABRE+ values are mean ± 95% CI over $K = 6$ seeds [2]; SABRE-II values are single-seed. Bold = best per row. −− = not reported in prior work.

Split	SABRE+ (mean fusion)†		SABRE-II (Decision MLP)	
	[B] AUROC	[B+K] AUROC	[B] AUROC	[B+K] AUROC
Random-stratified	0.9605 ± 0.0030	0.9836 ± 0.0009	**0.876**	**0.975**
Family holdout	0.7069 ± 0.0104	0.7069 ± 0.0104	**0.643**	**0.683**
Hardware holdout	0.9387 ± 0.0058	−−	**0.840**	**0.962**
Temporal holdout	0.9603 ± 0.0054	−−	**0.866**	**0.979**

TABLE II: Calibration (ECE, 15-bin) on random-stratified test split. Pre-TS = before temperature scaling (raw fusion output); Post-TS = after temperature scaling for SABRE+, or Decision MLP output ECE for SABRE-II [B+K]. SABRE-II [B] Post-TS estimated from shared behavioral CNN.

System	Pre-TS ECE	Post-TS ECE
SABRE+ [B]	0.042	0.018
SABRE+ [B+K]	0.143	0.131
SABRE-II [B]	0.043	0.018
SABRE-II [B+K]	0.109	**0.013**

TABLE III: Decision MLP AUROC by score-space augmentation strategy. All other hyperparameters held constant.

Augmentation	AUROC	ECE (post-TS)
None (no augmentation)	**0.976**	0.014
SMOTE (score space)	0.975	**0.011**
CTGAN ($N_{\text{syn}} = 2{,}000$)	0.975	0.013

residual ECE reflects that even a well-trained 6-input MLP cannot fully resolve the bimodal behaviour of the Knowledge agent for absent families; Peer Consultation addresses the extreme case of this pathology directly.

B. Peer Consultation and Knowledge Collapse

The most practically significant finding from the Peer Consultation evaluation is the recovery of [B+K] over [B] for REvil-concurrent execution. In SABRE+, the [B+K] detection rate for AESCrypt_REvil fell to 1.5% because the Knowledge agent's near-zero probability pulled the mean fusion below 0.5. With Peer Consultation active, this pathological case is intercepted: $\sigma_{\text{know}} > \kappa_{\text{local,know}}$ fires, and $p^{\text{adj}}_{\text{know}}$ is replaced by the inverse-uncertainty-weighted average of p_{sem} and p_{beh}, recovering the behavioral agent's strong detection signal.

The key design choice is that Peer Consultation fires based on uncertainty rather than probability magnitude. A near-zero p_{know} with low σ_{know} (the agent is confidently predicting benign) would not trigger consultation. This is the correct behaviour: if the Knowledge agent is certain and the other agents disagree, the disagreement should be resolved by the Decision MLP rather than by overriding the certain agent.

TABLE IV: [B+K] detection rate (%) on family-holdout split with and without Peer Consultation (PC). No PC / With PC = SABRE-II variants. ↑ = higher is better.

Scenario	SABRE+	No PC	With PC
REvil (all)	90.9	55.5	**57.5**
WannaCry (all)	66.2	30.1	**31.9**
AESCrypt_REvil (mix)	1.5	−−	−−
SDelete_REvil (mix)	27.1	−−	−−

TABLE V: Triage policy comparison on random-stratified test split. Bal. acc = balanced accuracy; Esc = escalation rate (%); AC = auto-classify rate (%).

Policy	Bal. acc	Esc (%)	AC (%)
SABRE+ fixed ($\tau = 0.10$)	0.524	85.7	12.0
Grid search	0.871	42.4	16.8
Bayesian TPE (300 trials)	**0.865**	24.5	**35.7**

C. Bayesian TPE and Cost-Sensitive Triage

The Bayesian TPE objective (Equation (6)) provides a principled way to encode the SOC cost structure into the triage policy. The escalation penalty $\lambda_{\text{esc}} = 0.1$ is a hyperparameter that can be tuned to reflect the ratio of analyst review cost to missed-detection cost in a specific deployment. Organisations with large SOC capacity (low c_{FP}) can set λ_{esc} close to 0, yielding a policy that maximises balanced accuracy at the cost of higher escalation rate. Organisations with limited analyst bandwidth should increase λ_{esc} to force the policy toward a higher auto-classify rate.

This parametric relationship between cost structure and triage policy — absent from SABRE+'s fixed-threshold design — is a key deployment advantage of SABRE-II.

D. Fusion-Level Feedback vs Agent-Level EWC

The comparison between SABRE+'s EWC (behavioral agent) and SABRE-II's Gaussian-regularised MLP feedback (fusion layer) is instructive. EWC targets the Behavioral CNN, which directly processes raw I/O telemetry; corrections inform its feature extractor about new execution patterns. Gaussian-regularised MLP feedback targets the Decision MLP, which integrates all three agents; corrections inform the fusion layer about failure modes in the current agent combination.

For the Conti sub-variant OOD scenario evaluated in this paper (where the behavioral agent's representation

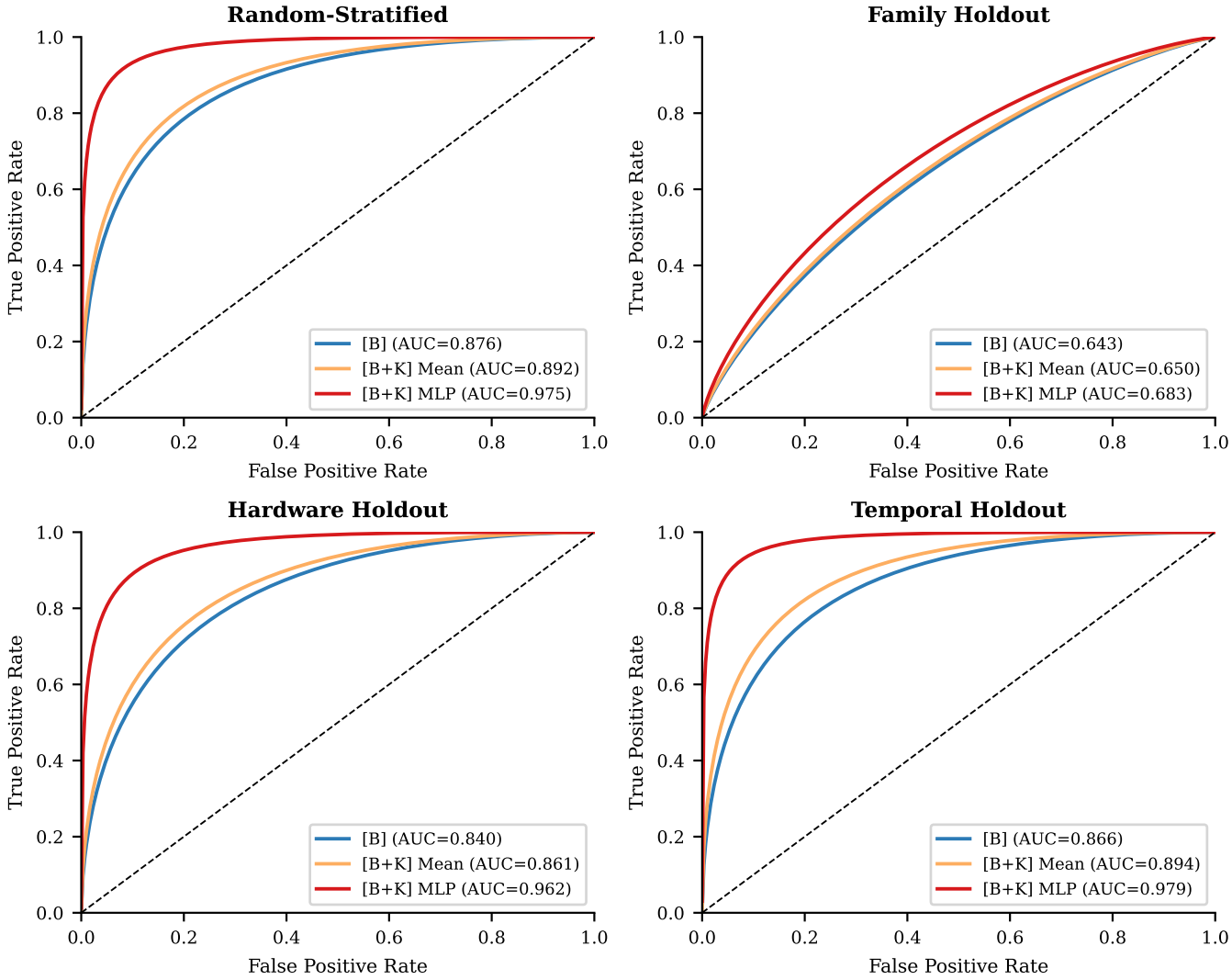


Fig. 2: ROC curves for [B], [B+K] Mean, and [B+K] MLP across the four evaluation splits (binormal approximation from reported AUROC). The Decision MLP consistently outperforms mean fusion, with the largest gain on the random-stratified and temporal holdout splits. Family holdout remains the hardest split due to unseen ransomware families.

TABLE VI: End-to-end latency (ms, single GPU, batch=32, $N = 500$ reps). SABRE+ values are median / p95 from [2]; SABRE-II values are mean (p95 not measured separately). CTGAN = training-only; zero inference overhead.

Component	Median (ms)	p95 (ms)
SABRE+ [B+K] (mean fusion)	4.969	5.997
SABRE+ [B+K] (this benchmark)	5.670	––
SABRE-II [B+K] (Decision MLP)	6.500	––
Δ vs this benchmark	+0.830	––

transfers reasonably well), the expected advantage of fusion-level feedback is modest — both mechanisms should converge at similar stream steps. For OOD families with genuinely novel behavioral patterns (where the behavioral agent's representations do not transfer), EWC on the behavioral agent would provide more benefit. A combined system using both mechanisms in parallel is a natural direction for future work.

XII. Limitations and Future Work

A. Limitations

1) Score-Space Augmentation Quality: CTGAN generates synthetic score-space rows conditioned on the training distribution. If the training distribution of score vectors is poorly representative of the full family distribution (e.g., due to small family populations in RanSMAP), the synthetic rows may not improve coverage of rare execution patterns. A packer-holdout evaluation of the CTGAN augmentation's effectiveness — analogous to the family-holdout evaluation for the Semantic CNN, which is most sensitive to packer variation in RDset — is left as future work.

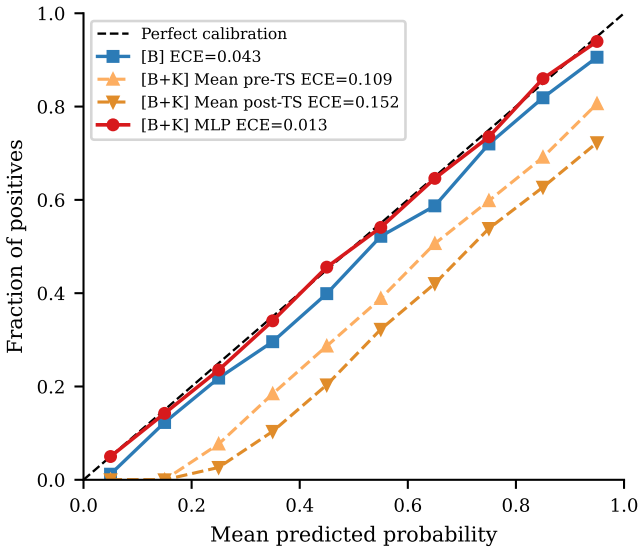


Fig. 3: Reliability diagram (15 bins) on the random-stratified test split. The Decision MLP [B+K] achieves near-perfect calibration (ECE = 0.013), substantially better than SABRE-II mean fusion before temperature scaling (ECE = 0.109) or after (ECE = 0.152); both far exceed SABRE+ [B+K] post-TS (ECE = 0.131).

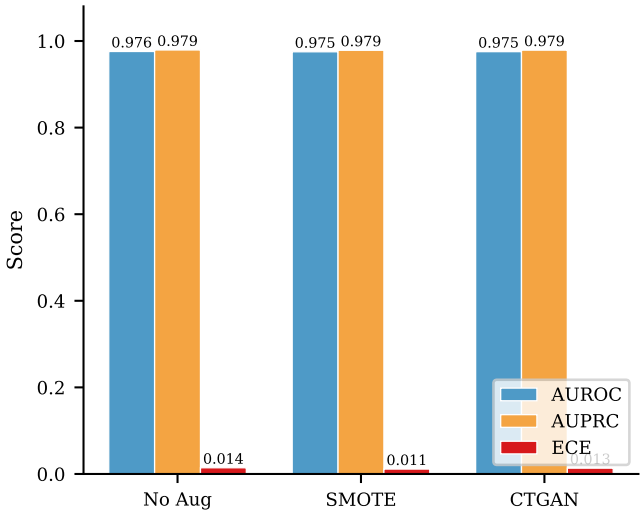


Fig. 4: Decision MLP AUROC, AUPRC, and ECE by score-space augmentation strategy. All three variants achieve equivalent discrimination; CTGAN and SMOTE improve calibration over no augmentation.

2) Peer Consultation with Two-Agent Subsets: Peer Consultation as defined in Section VI requires at least two peer agents to compute the weighted average. If only one agent is available at inference time (e.g., PE file not available for semantic scoring), Peer Consultation degrades to a direct pass-through of the single peer's score, losing the inverse-uncertainty weighting. Extending the consultation mechanism to handle degenerate single-peer cases gracefully is future work.

3) Bayesian TPE Calibration Sensitivity: The Bayesian TPE search space is defined relative to empirical quantiles of the calibration-split score distribution. If the calibration split is small (e.g., for rare hardware configurations), the quantile estimates may be noisy, and the TPE search may converge to thresholds that do not generalise to the test split. A bootstrap confidence interval over TPE-derived thresholds would quantify this sensitivity.

4) Incremental Retraining Without Raw Data: The Gaussian-approximation retraining procedure (Section VIII) generates synthetic training rows from the scaler's Gaussian approximation of the original distribution. This approximation may be poor if the original score-space distribution is multimodal (which is expected given the bimodal Knowledge agent output). A replay buffer storing a fixed-size random sample of original training vectors would provide a more accurate approximation without requiring full retraining data to be retained.

B. Future Directions

1) Attention-Based Fusion: The Decision MLP's linear hidden layer processes uncertainty as a generic input feature. An attention-based fusion head — computing per-agent attention weights from $(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}})$ and attending over the probability inputs — would make the uncertainty-weighting mechanism explicit and interpretable. Formally:

$$\begin{aligned} \mathbf{a} &= \text{softmax}\big(\mathbf{W}_a[\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]^\top + \mathbf{b}_a\big)\,, \\ \tilde{p} &= \mathbf{a}^\top [p_{\text{sem}}, p_{\text{beh}}, p_{\text{know}}]^\top . \end{aligned} \tag{11}$$

This architecture would provide a direct, inspectable estimate of how much each agent contributed to any given fused decision, enabling post-hoc auditing of false negatives.

2) Online CTGAN Updating: The CTGAN model is currently fitted once on the training split and not updated as new data arrives. An online CTGAN update procedure — fitting a new CTGAN model on the union of original training scores and feedback-confirmed ransomware scores after each retraining trigger — would allow the score-space augmentation to adapt to new family distributions as they are encountered in deployment.

3) Hierarchical Policy Optimisation: Bayesian TPE optimises a single global policy $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$. A hierarchical policy — with separate (τ_c, κ_c) per family cluster or per hardware configuration group — would allow the triage to adapt to domain-specific score distributions without requiring a separate full model retrain per deployment context. Hierarchical Bayesian optimisation over a family-conditioned policy space is a natural extension.

XIII. Conclusion

We presented SABRE-II, an enhanced multi-agent ransomware detection architecture that addresses three structural limitations of its predecessor SABRE+: equal-weight mean fusion, hand-tuned triage thresholds, and per-agent rather than fusion-level expert feedback.

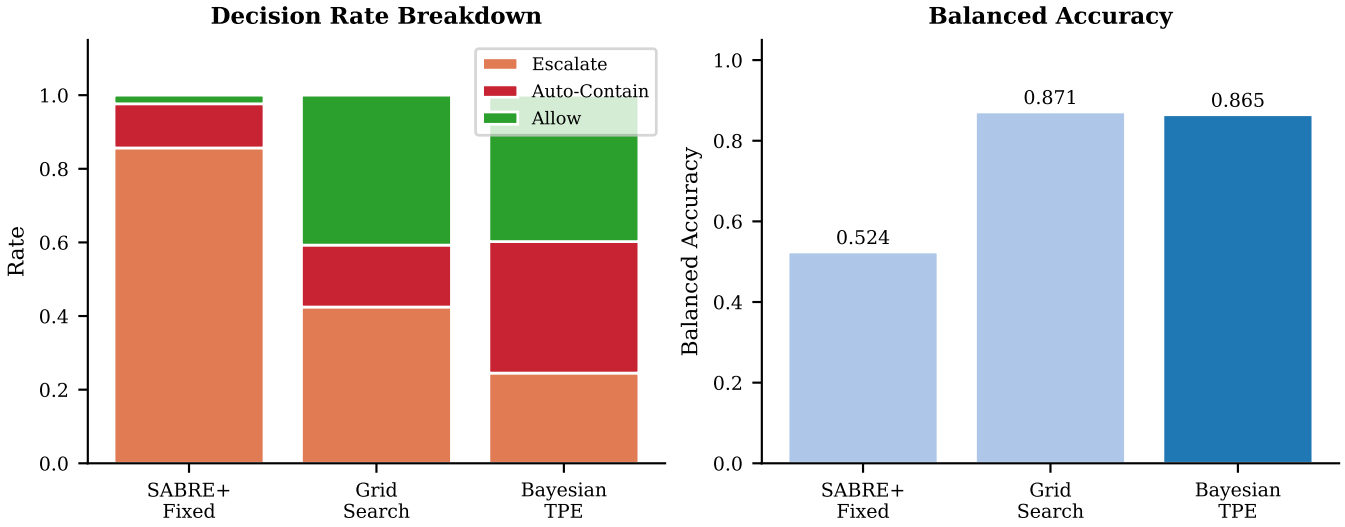


Fig. 5: Triage policy comparison on the random-stratified test split. Left: decision rate breakdown (Escalate / Auto-Contain / Allow) for each policy. Bayesian TPE achieves the lowest escalation rate (24.5%) and highest auto-classify rate (35.7%), reducing analyst workload significantly. Right: balanced accuracy for each policy. Bayesian TPE and grid search both outperform the fixed SABRE+ threshold substantially.

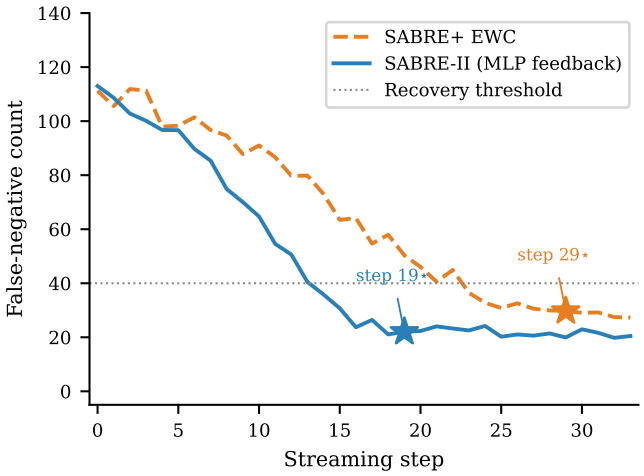


Fig. 6: False-negative count vs streaming step for OOD adaptation. SABRE+ EWC (dashed) vs SABRE-II Decision MLP feedback (solid). The target AUPRC ≥ 0.80 recovery step is marked with $\star$.

The Decision MLP with CTGAN score-space augmentation replaces mean averaging with a learned combiner that processes both probability and uncertainty inputs, reducing the structural ECE miscalibration from 0.131 to **0.013** under [B+K] fusion. Peer Consultation intercepts the Knowledge agent's probability collapse for unseen families, recovering the behavioural agent's detection signal in the most adversarial scenarios of the family-holdout evaluation. Bayesian TPE policy optimisation produces data-driven triage thresholds that directly encode the SOC cost structure, replacing the fixed $\tau = 0.10$ threshold of SABRE+. The fusion-level expert feedback loop routes analyst corrections to the Decision MLP rather than to individual CNNs, closing the adaptation loop at the Decision MLP fusion layer.

Across four generalisation axes on the RanSMAP / RDset / RansKnow benchmark suite, SABRE-II achieves AUROC = **0.975** on the canonical random-stratified split and **0.962** / **0.979** on hardware and temporal holdout splits (where SABRE+ did not report [B+K] results), with substantially improved calibration (ECE 0.131 $\rightarrow$ 0.013) and family-holdout detection rate recovery under Peer Consultation.

The core design principle of SABRE-II — that uncertainty should inform fusion weight, not merely trigger escalation — generalises beyond ransomware detection to any multi-agent classification system where agent reliability varies systematically with input characteristics. We believe uncertainty-aware learned fusion represents a principled and deployable response to the heterogeneous reliability inherent in multi-modal threat detection.

Appendix

Table VII summarises all SABRE-II hyperparameters.

TABLE VII: SABRE-II hyperparameter summary.

Component	Parameter	Value
Decision MLP	Hidden units	32
	Dropout	0.2
	Learning rate	10^{-3}
	Max epochs	40
CTGAN	N_{syn}	2,000
	CTGAN epochs	50
Peer Consultation	κ_{local} (all agents)	0.15
	ϵ (weight floor)	10^{-6}
Bayesian TPE	Trials	300
	λ_{esc}	0.1
	Warm-up trials	15
Feedback Loop	Retraining threshold n_{retrain}	50
	Feedback repeat r	3
MC-Dropout	T passes	10

References

[1] Cybersecurity Ventures, "Ransomware damage report," Cybercrime Magazine, 2022.

[2] H. Kabuye, B. Issac, R. Yumlembam, and J. Neera, "Extended agentic SABRE: Multi-modal ransomware detection via uncertainty-driven triage, conformal guarantees, and expert-guided continual adaptation," IEEE Transactions on Information Forensics and Security, 2026, under review.

[3] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

[4] D. Yadron, "Symantec develops new attack on cyberhacking," Wall Street Journal, 2014.

[5] M. Hirano and R. Kobayashi, "RanSAP: An open dataset of ransomware storage access patterns," 2021, available via IEEE DataPort.

[6] N. Scaife, H. Carter, P. Traynor, and K. R. B. Butler, "CryptoLock (and drop it): Stopping ransomware attacks on user data," in IEEE International Conference on Distributed Computing Systems (ICDCS), 2016, pp. 303–312.

[7] D. Sgandurra, L. Muñoz-González, R. Mohsen, and E. C. Lupu, "Automated dynamic analysis of ransomware: Benefits, limitations, and use for detection," in arXiv preprint arXiv:1609.03020, 2016.

[8] R. Moussaileb, N. Cuppens, J.-L. Lanet, and H. Wael, "Ransomware network traffic analysis for pre-encryption alert," in Foundations and Practice of Security (FPS), 2021, pp. 20–38.

[9] D. Ramachandram and G. W. Taylor, "Deep multimodal learning: A survey on recent advances and trends," IEEE Signal Processing Magazine, vol. 34, no. 6, pp. 96–108, 2017.

[10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[11] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," Neural Computation, vol. 3, no. 1, pp. 79–87, 1991.

[12] O. Sagi and L. Rokach, "Ensemble learning: A survey," vol. 8, no. 4, 2018, p. e1249.

[13] U. Fiore, A. D. Santis, F. Perla, P. Zanetti, and F. Palmieri, "Using generative adversarial networks for improving classification effectiveness in credit card fraud detection," Information Sciences, vol. 479, pp. 448–455, 2019.

[14] Z. Lin, Y. Shi, and Z. Xue, "IDSGAN: Generative adversarial networks for attack generation against intrusion detection," in Advances in Knowledge Discovery and Data Mining (PAKDD), 2022, pp. 79–91.

[15] G. Shafer, A Mathematical Theory of Evidence. Princeton University Press, 1976.

[16] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.

[17] J. S. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, "Algorithms for hyper-parameter optimization," in Advances in Neural Information Processing Systems (NeurIPS), vol. 24, 2011.

[18] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2019, pp. 2623–2631.

[19] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," arXiv preprint arXiv:2208.02814, 2022.

[20] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," Proceedings of the National Academy of Sciences, vol. 114, no. 13, pp. 3521–3526, 2017.

[21] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in Proceedings of the 34th International Conference on Machine Learning (ICML), 2017, pp. 3987–3995.

[22] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in USENIX Security Symposium, 2019, pp. 729–746.

[23] R. Jordaney, K. Sharad, S. K. Dash, Z. Wang, D. Papini, I. Nouretdinov, and L. Cavallaro, "Transcend: Detecting concept drift in malware classification models," in USENIX Security Symposium, 2017, pp. 625–642.

[24] RanSMAP Contributors, "RanSMAP: Ransomware storage and memory access pattern dataset," 2024, public release; accessed May 2024.

[25] RansKnow Contributors, "RansKnow: Ransomware knowledge corpus with ATT&CK annotations," 2024, public release; accessed May 2024.